

Supplementary Material: Uncollapsed and Metrically-Scaled 3D Reconstruction from Monocular Endoscopic Video via Pseudo-3D Heightfields

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This supplementary material provides additional details on the experimental configuration (Section A), per-frame pose error analysis (Section B), PCA ellipsoid parameters (Section C), algorithm pseudocode (Section D), extended ablation study (Section E), additional qualitative visualizations (Section F), per-sequence stereo laparoscopic results (Section G), newly captured sequences (Section H), and the homography branch-veto evaluation (Section I). Evaluation scripts, the pipeline of record, and the result JSONs used in the main text are at <https://github.com/liranyang123456-commits/re3d-heightfield>.

Notation Reference

A. Detailed Experimental Configuration

A.1 Hardware and Software Environment

All experiments were conducted on the following platform:

- **CPU:** Intel Core i9-14900K (32 cores, 5.8 GHz turbo)
- **GPU:** NVIDIA GeForce RTX 5090 (24 GB VRAM, CUDA 12.4)
- **RAM:** 64 GB DDR5-5600
- **OS:** Ubuntu 22.04 LTS / Windows 11
- **Python:** 3.9 (Open3D 0.19.0, OpenCV 4.13.0; PyTorch 2.x for the stereo variant)

A.2 Hyperparameter Configuration

Table 2 provides the complete hyperparameter configuration of our method.

A.3 Baseline Implementation Details

Table 3 provides the implementation details for each baseline method.

B. Per-Frame Pose Error Analysis

Table 4 provides the per-method pose statistics against the independent `solvePnP` chessboard reference. ATE is the position RMSE after Sim3 alignment; RPE-t is the mean relative translation error between consecutive frame pairs (alignment-free); Scale is the Sim3 scale factor.

Key observations..

1. **Metric scale is the decisive difference.** Ours recovers a near-ideal scale factor ($1.11\times$); every baseline drifts, from $3.7\times$ (ORB-SfM) up to $647\times$ (DUST3R). Even the best feed-forward method (VGGT) drifts by $91\times$, which precludes metric measurement despite its accurate trajectory shape.
2. **Position accuracy is competitive.** VGGT attains the lowest ATE (3.02 mm); Ours (3.89 mm) is within 0.9 mm and comparable to ORB-SfM (3.79 mm), while DUST3R, MAST3R, Reloc3R, and COLMAP exceed 20 mm.
3. **No local drift.** Relative translation errors are comparable across all methods (2.2–3.3 mm per pair), indicating that the differences lie in global scale and long-range consistency rather than per-frame increments.
4. **Rotation is not discriminative on this sequence.** The reference relative rotation is only $\approx 0.3^\circ$ per frame; the learned and SfM baselines achieve relative rotation errors of 0.4° – 5.1° per pair. The heightfield-ICP relative rotation (16.1° per pair) is approximate because it is estimated in the pseudo-metric heightfield embedding rather than in metric camera

space (see Limitations in the main text); on this translation-dominant sequence it does not affect the fused positions, scale, or the reconstruction.

C. PCA Ellipsoid Parameters and Shape Analysis

Table 5 provides the complete PCA eigenvalue analysis for all methods, including the raw eigenvalues, eigenvalue ratios, and spatial extent along each principal axis.

Shape classification criteria.. We classify point cloud shapes based on the λ_3/λ_1 ratio:

- **Normal 3D:** $\lambda_3/\lambda_1 \geq 0.25$ — the point cloud exhibits balanced 3D distribution.
- **Flat:** $0.15 \leq \lambda_3/\lambda_1 < 0.25$ — the point cloud collapses along one principal axis.
- **Cone:** $\lambda_3/\lambda_1 < 0.15$ — the point cloud is severely elongated, indicating near-degenerate geometry.

Interpretation of extent ratios.. The spatial extent along the three principal axes further corroborates the PCA analysis:

- **ORB-SfM:** Extent $135.3 \times 59.8 \times 199.8$ shows a pronounced elongation along the third axis, consistent with cone-shaped degeneracy.
- **DUS3R:** The PCA ratio (0.09) reveals a cone-shaped point distribution; its apparent spatial spread arises from the extreme scale drift ($647\times$, Table 4) rather than genuine 3D structure.
- **Ours:** Extent $191 \times 200 \times 193$ mm with ratio $1.00 : 1.05 : 1.01$ is nearly isotropic, confirming balanced 3D distribution at the correct metric scale.

D. Algorithm Pseudocode

We provide the complete pseudocode for the three main stages of our pipeline.

Algorithm 1 Pseudo-3D Heightfield Mesh Construction

Require: Image I_t , chessboard mask M_t , threshold $\tau\%$, cell size c_{cell}

Ensure: Heightfield mesh $\mathcal{H}_t = (\mathcal{V}_t, \mathcal{F}_t)$

- 1: $g_t \leftarrow \|\nabla I_t\|$ {Gradient magnitude}
 - 2: $\tau_g \leftarrow \text{Perc}_{\tau\%}(g_t)$ {Percentile threshold}
 - 3: $\hat{g}_t(u, v) \leftarrow g_t(u, v)$ if $g_t(u, v) \geq \tau_g$, else 0
 - 4: $\tilde{g}_t \leftarrow \hat{g}_t \odot M_t$ {Apply chessboard mask}
 - 5: **for** each grid vertex (u, v) with step c_{cell} **do**
 - 6: $h(u, v) \leftarrow z_s \cdot \tilde{g}_t(u, v)/255$ {Height assignment}
 - 7: $\mathcal{V}_t \leftarrow \mathcal{V}_t \cup \{(u, v, h(u, v))\}$
 - 8: **end for**
 - 9: $\mathcal{F}_t \leftarrow \text{RegularGridTriangulation}(\mathcal{V}_t)$
 - 10: **return** $\mathcal{H}_t$
-

E. Extended Ablation Study

E.1 Cell Size Sensitivity

Table 6 evaluates the impact of the heightfield grid step c_{cell} on reconstruction quality, with the camera poses held fixed.

All grid steps produce Normal 3D reconstructions, but the finer grids ($c_{\text{cell}} \leq 3$) achieve the highest shape ratios (≥ 0.39), because a denser heightfield preserves more of the gradient structure that drives both matching and fusion. Coarser grids ($c_{\text{cell}} \geq 4$) remain Normal 3D but lose 0.05–0.11 of shape ratio as the heightfield undersamples the tissue-surface topology. We therefore use $c_{\text{cell}}=3$ as the default; note that this is the grid step of the heightfield mesh, which the main text denotes c_{cell} .

E.2 Search Radius Sensitivity

Table 7 evaluates the dense-matching search radius r using the paper’s evaluation pipeline with the original meshes and heightfield-ICP poses.

The reconstruction is *insensitive* to r over $[10, 40]$: the pose-predicted search window and the ± 2 px epipolar band already localise each candidate to a few

Algorithm 2 Heightfield-Based Pose Estimation

Require: Heightfield meshes $\mathcal{H}_t, \mathcal{H}_{t+1}$ (as point clouds)

Ensure: Relative transformation $\mathbf{T}_{t+1,t}$

```
1: Estimate normals; compute FPFH features for both clouds
2:  $\mathbf{T}_{\text{pca}} \leftarrow \text{PCAAlignment}(\mathcal{H}_t, \mathcal{H}_{t+1})$  {principal-axis alignment}
3: if residual( $\mathbf{T}_{\text{pca}}$ )  $> 0.1 \cdot \text{scale}$  then
4:    $\mathbf{T}_0 \leftarrow \text{RANSACFeatureMatching}(\mathcal{H}_t, \mathcal{H}_{t+1}, \text{FPFH}, 5d)$  {fallback}
5: else
6:    $\mathbf{T}_0 \leftarrow \mathbf{T}_{\text{pca}}$ 
7: end if
8:  $\mathbf{T} \leftarrow \mathbf{T}_0$ 
9: while not converged do
10:   $\mathcal{N}(i) \leftarrow$  nearest-neighbour of  $\mathbf{T}\mathbf{p}_i$  in target
11:   $\mathbf{T} \leftarrow \arg \min_{\mathbf{T}} \sum_i \|\mathbf{T}\mathbf{p}_i - \mathbf{q}_{\mathcal{N}(i)}\|^2$  {point-to-point}
12: end while
13: fit  $\leftarrow$  ICP fitness( $\mathbf{T}$ )
14: if fit  $< 0.1$  then
15:   re-initialise from RANSAC / centroid alignment; retry
16: end if
17: return  $\mathbf{T}_{t+1,t} \leftarrow \mathbf{T}$ 
```

pixels, so enlarging the window adds no new matches on this benchmark, and shrinking it to 10px does not remove correct ones. The default $r=25$ provides margin for the larger inter-frame displacements of free-hand motion without measurable cost.

E.3 Note on ICP Configuration

The registration stage uses single-scale point-to-point ICP with an adaptive, scale-derived correspondence distance and the FPFH/PCA coarse initialisation described in the main text; a multi-resolution ICP pyramid is not part of the proposed pipeline. The pose-source ablation in the main paper (Table 5) isolates

Algorithm 3 Dense Epipolar Matching and Metric Fusion

Require: Heightfield meshes $\{\mathcal{H}_t\}$, poses $\{\mathbf{T}_t\}$, intrinsics $\mathbf{K}$

Ensure: Fused point cloud $\mathcal{P}$

```
1:  $\mathcal{C} \leftarrow \emptyset$  {Correspondence set}
2: for each frame pair  $(t, t+1)$  do
3:    $\mathbf{F} \leftarrow \mathbf{K}^{-\top}[\mathbf{t}]_{\times} \mathbf{R} \mathbf{K}^{-1}$  {Fundamental matrix}
4:   for each valid grid point  $(u, v)$  in  $\mathcal{H}_t$  do
5:      $\mathbf{f} \leftarrow \text{Extract7DFeature}(u, v, \mathcal{H}_t)$ 
6:      $(u', v') \leftarrow \mathbf{T}_{t+1, t} \cdot (u, v, h)$  {RT prediction}
7:      $k^* \leftarrow \arg \min_{k \in W_r} \|\mathbf{f} - \mathbf{f}_k\|_2$  s.t.  $|\mathbf{u}'^{\top} \mathbf{F} \mathbf{u}| < \epsilon_{\text{epi}}$ 
8:     if CrossCheck( $k^*$ ) and DepthFilter( $k^*$ ) then
9:        $\mathcal{C} \leftarrow \mathcal{C} \cup \{(u, v) \leftrightarrow (u_{k^*}, v_{k^*})\}$ 
10:    end if
11:  end for
12: end for
13:  $\mathcal{P}_{\text{raw}} \leftarrow \text{Triangulate}(\mathcal{C}, \mathbf{K}, \{\mathbf{T}_t\})$  {Linear triangulation}
14:  $\mathcal{P}_{\text{filtered}} \leftarrow \text{MultiViewConsistency}(\mathcal{P}_{\text{raw}}, \tau_{\text{reproj}})$ 
15:  $\mathcal{P} \leftarrow \text{VoxelDedup}(\mathcal{P}_{\text{filtered}}, 1.0 \text{ mm}) \rightarrow \text{OutlierRemoval}(k=20, \sigma=2.0)$ 
16: return  $\mathcal{P}$ 
```

the contribution of this registration stage.

E.4 Learned Stereo Variants (V2–V4)

Table 8 reports the five learned stereo variants analysed in Section 4.7 of the main text. V2 is a lightweight custom encoder (4.76M parameters) with channel-wise dot-product cost volume; V2+ adds a refinement head; V3 scales the encoder to 26.5M parameters; V3s is a reduced V3; V4 couples a pretrained ResNet encoder with group-wise correlation [4]. All variants are trained on EndoNeRF (174 samples) for up to 120 epochs and evaluated in-domain (43 validation samples) and zero-shot on SCARED (15 keyframes, no fine-tuning).

The Pareto frontier is flat in-domain between V2 and V2+ (5–6M param-

ters), while accuracy collapses beyond 23M parameters because the 174-sample training set cannot sustain larger encoders. Zero-shot SCARED errors are two orders of magnitude larger than in-domain errors for every variant, which is why the main text uses the calibrated variant fine-tuned with the heightfield loss rather than a zero-shot checkpoint.

F. Additional Qualitative Visualizations

F.1 PCA Ellipsoid Comparison

Fig. 1 shows the PCA ellipsoid visualization for each method, illustrating the 3D distribution of the reconstructed point clouds. The ellipsoid axes are scaled by the square root of the corresponding PCA eigenvalue, providing an intuitive representation of the point cloud’s shape.

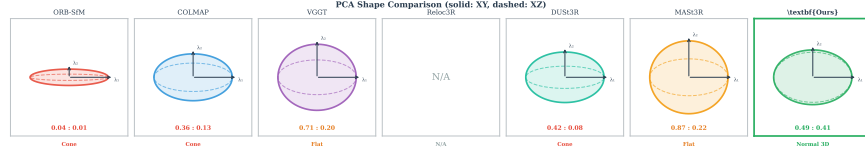


Figure 1: PCA ellipsoid comparison. Each ellipsoid’s axes are proportional to $\sqrt{\lambda_i}$, the square root of the corresponding eigenvalue. Ours produces a near-spherical ellipsoid ($\lambda_3/\lambda_1 = 0.42$), while baselines produce highly elongated (cone) or flattened (flat) ellipsoids. Best viewed in color.

F.2 Heightfield Mesh Examples

To illustrate the pseudo-3D heightfield mesh construction, we describe the intermediate results at each stage of the pipeline:

1. **Input image:** A 640×358 endoscopic frame (downsampled from the native 1280×720) containing the chessboard marker, surrounding tissue, and surgical instruments.
2. **Gradient magnitude:** The Sobel gradient map highlights structural edges (chessboard boundaries, instrument contours) while producing relatively uniform responses in specular regions.

3. **Truncated gradient:** After percentile thresholding ($\tau_{\%} = 68$), weak gradients are zeroed, producing a sparse but reliable height field.
4. **Chessboard mask:** The binary mask from corner detection delineates the reliable reconstruction region.
5. **Masked heightfield:** The product of truncated gradient and chessboard mask, ready for UVH mesh construction.
6. **UVH mesh:** The 2.5-D triangulated surface with $|\mathcal{V}_t| \approx 25,680$ vertices at $c_{\text{cell}}=3$. The 11,735 figure in the main-text chessboard table is the fused point count after triangulation and filtering, not the per-frame mesh size.

F.3 Failure Cases

We identify two failure modes of the proposed method:

Partial chessboard occlusion.. When the chessboard marker is partially occluded by a surgical instrument, corner detection fails to detect the full corner set, resulting in a reduced mask that covers only the visible portion of the board. In such frames, the heightfield mesh is constructed over a smaller region, potentially leading to less constrained ICP registration. The quality gating mechanism rejects registrations with low fitness, preventing corrupted poses from entering the reconstruction.

Large inter-frame displacement.. When the camera moves rapidly between consecutive frames (displacement > 25 px), the dense matching search radius may be insufficient to find correct correspondences. This can be mitigated by increasing r at the cost of higher computational expense and increased risk of false matches, or by using feature-based RANSAC as a fallback initializer.

G. Per-Sequence Stereo Laparoscopic Results

Fig. 2 shows the per-sequence δ_1 accuracy comparison, and Table 9 provides the per-sequence depth accuracy and shape classification for all 22 sequences.

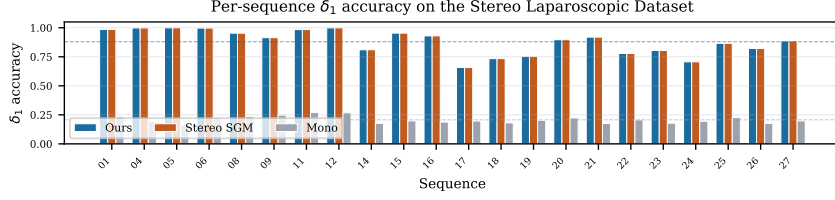


Figure 2: Per-sequence δ_1 accuracy on the Stereo Laparoscopic Dataset (22 sequences, five endoscopes). Ours and Stereo SGM share the same SGM disparity source and therefore attain identical per-sequence depth accuracy; the heightfield contributes a per-pixel confidence map rather than a depth change. Dashed lines mark the per-method mean.

Key observations..

1. Ours and Stereo SGM attain identical AbsRel and δ_1 on all 22 sequences, because the depth source is the same SGM disparity; the contribution of the heightfield on this dataset is the per-pixel confidence map, not a depth-accuracy gain (see main text).
2. Mono consistently produces Cone shapes ($\lambda_3/\lambda_1 < 0.15$) on 21/22 sequences, confirming that gradient-only inverse-depth provides insufficient geometric structure.
3. The per-sequence results demonstrate generalization across five different endoscopes and diverse surgical scenes, with no re-tuning.

H. Newly Captured Endoscopic Sequences (Detailed Results)

This section provides the full per-session metrics behind Section 4.8 of the main text. Four sequences were captured with the left channel of a different stereo-endoscope rig (1280×720, 30 fps, calibrated intrinsics $f_x=1210.7$, $f_y=1194.6$ px, distortion-corrected). Ground-truth poses come from a 12×9 chessboard with 3.0mm squares (PnP, reprojection RMSE <0.7 px). Planar-PnP flip outliers are removed before ATE and before the GT-rotation column; on P1, residual steps $\geq 60^\circ$ are also excluded from that column because the neighbour-median filter still leaves 180° IPPE flips. Unfiltered means (16.4°/5.1°/32.3°/11.1°) are flip-inflated and are not used. The pipeline runs with self-estimated poses on

all frames. Pose-valid frames are 24%, 28%, 26%, and 12% (V1, V2, P1, V3; $n_{\text{GT}}/n_{\text{frames}}$).

H.1 Full per-session metrics

Table 10 reports shape, pose, and corner-depth metrics for the proposed method on all four sequences.

H.2 Rotation-stratified position error

Table 11 breaks the per-frame position error down by the reference relative-rotation magnitude.

H.3 Marker-free segments

On frames where the chessboard is not visible, the pipeline falls back to a contour-based ROI mask (largest contours of the truncated gradient map) and continues to reconstruct. This fallback was exercised on all four sequences (the board is visible on 12–28% of frames; V3 is 12%), and the fused clouds remain spatially coherent through the marker-free stretches; however, because no ground truth exists on those stretches, the evaluation above is restricted to board-visible frames, and the marker-free segments are assessed qualitatively.

H.4 Multi-scale boards (N2)

The recordings also contain two smaller calibration targets. Table 12 lists valid-frame counts and the camera-to-board working distance of the 12×9–3 mm trajectory board. The 9×6–5 mm board appears on at most 19 frames (P1) and the 9×8–8 mm board on at most one frame per sequence. These counts are not enough for a second Sim3 ATE protocol, so the paper uses only the 12×9–3 mm poses. Mean standoff on that board is 177–216 mm.

I. Homography Branch-Veto Evaluation

This section quantifies the branch-selection behaviour behind Section 3.3 of the main text. On the chessboard benchmark (first 50 frames, calibrated

$f=2304$ px), we extract SIFT correspondences for frame pairs at strides $s \in \{1, 2, 4, 8\}$, estimate the RANSAC homography, and keep the two cheirality-satisfying branches of the decomposition. For each pair we determine the *reference-consistent branch*—the branch whose translation direction agrees more with the stereo-verified reference motion—and measure how often each selection rule picks it. Three rules are compared: (i) the heightfield normal veto with the true board normal (an upper bound on the criterion), (ii) the veto with the constant rig-dominant normal used by the pipeline of record, and (iii) the standard max-cheirality heuristic.

Two observations follow. First, the veto criterion is sound: given a faithful surface normal, it selects the reference-consistent branch as often as or more often than cheirality counting, with the advantage concentrated at wide strides where branch confusion is most damaging. Second, the constant rig prior is a weak surrogate for the per-frame heightfield normal on this bench-top sequence (the board inclination varies over the sweep), which is why the pipeline couples the veto with the multi-stride bundle adjustment rather than trusting any single decomposition: residual branch errors behave as transient outliers that the loop-closed optimisation rejects. Improving the per-frame normal estimate (e.g., from the learned disparity in the stereo variant) directly strengthens the veto, as Table 13 indicates.

References

- [1] G. Bradski, “The OpenCV library,” *Dr. Dobb’s J. Software Tools*, 2000.
- [2] J. Sturm *et al.*, “A benchmark for the evaluation of RGB-D SLAM systems,” in *IROS*, 2012, pp. 573–580.
- [3] S. Choi *et al.*, “Robust reconstruction of indoor scenes,” in *CVPR*, 2015, pp. 5556–5565.
- [4] X. Guo, K. Yang, W. Yang, X. Wang, and H. Li, “Group-wise correlation stereo network,” in *CVPR*, 2019, pp. 3273–3282.

Table 1: Mathematical symbols used in the main text.

Symbol	Meaning
I_t, N	Frame t of the monocular video; total frame count
$\mathbf{T}_t = [\mathbf{R}_t \mathbf{t}_t]$	Camera pose $\in \text{SE}(3)$, world-to-camera
$\mathbf{C}_t = -\mathbf{R}_t^\top \mathbf{t}_t$	Camera centre in world coordinates
$\mathbf{K}, f_x, f_y$	Intrinsic matrix; focal lengths in pixels
$(u, v), H \times W$	Pixel coordinates; image rows $\times$ columns
$g_t, \hat{g}_t, \tilde{g}_t$	Gradient magnitude; after percentile truncation; after ROI mask
$\tau_g, \tau_\%$	Gradient threshold; its percentile (68)
M_t	Binary region-of-interest mask
$h(u, v), z_s$	Pseudo-height; height scale $0.15 \max(H, W)$
$\mathcal{H}_t = (\mathcal{V}_t, \mathcal{F}_t)$	Heightfield mesh: vertices, facets
c_{cell}	Mesh cell size (3 px)
$\mathbf{n}_{hf}, \bar{\mathbf{n}}_{hf}$	Per-vertex normal field; its median (operative normal)
$\mathbf{H}, \tilde{\mathbf{H}}$	Homography; normalised homography $\mathbf{K}^{-1} \mathbf{H} \mathbf{K}$
$\mathbf{R}_{ij}, \mathbf{t}_{ij}$	Relative rotation / translation between frames i, j
$\mathbf{n}_i, d_i$	Plane normal; camera-to-plane distance
k^*	Branch index selected by the normal veto
$\mathcal{E}, S_{\text{max}}$	Multi-stride co-visibility edges; max stride (3–5)
$\xi_i \in \mathfrak{se}(3)$	Twist coordinates $(\boldsymbol{\omega}^\top, \mathbf{v}^\top)^\top$ of pose i
$\mathbf{V}(\boldsymbol{\omega})$	Left $\text{SO}(3)$ Jacobian
$\rho_{\text{Huber}}, \delta_H$	Robust penalty; its threshold (3.0 px)
d_{min}	Cheirality barrier depth (10 mm)
$s_{\text{sq}}, p_{\text{sq}}$	Physical square pitch (3.0/5.0 mm); its pixel pitch
d_t	Standoff distance $f s_{\text{sq}} / p_{\text{sq}}$
$\mathbf{f}(u, v)$	7-D feature: position, height, normal, grey value
r, ϵ_{epi}	Dense-match search radius (25 px); epipolar band (2.0 px)
$\mathbf{F}$	Fundamental matrix $\mathbf{K}^{-\top} [\mathbf{t}]_{\times} \mathbf{R} \mathbf{K}^{-1}$
$\pi(\cdot)$	Perspective projection
$\lambda_1 \geq \lambda_2 \geq \lambda_3$	Point-cloud covariance eigenvalues
ATE, RPE	Absolute trajectory error; relative pose error
AbsRel, δ_1	Depth errors after the stated alignment
b, d_{disp}	Stereo baseline; disparity

Table 2: Complete hyperparameter configuration of the proposed method.

Component	Parameter	Value
Gradient	Percentile threshold $\tau\%$	68
	Height scaling z_s	adaptive
	Gradient method	Sobel 3×3
UVH Mesh	Cell size c_{cell}	3 px
	Triangulation	regular grid
ICP	Method	point-to-point
	Robust kernel	none (least-squares)
	Coarse alignment	PCA $\rightarrow$ FPFH+RANSAC
	Voxel downsample budget	20 000 pts
	Correspondence distance	scale-adaptive
Quality Gate	Primary	ICP fitness > 0.1 (relaxed to 0.3 after coarse)
	Optional (if landmarks)	$p_{95} < 5.0$ px
	Fallback	centroid-aligned ICP
Dense Matching	Search radius r	25 px
	Epipolar band ϵ_{epi}	2.0 px
	Cross-check tolerance	3 px
	Depth filter	$5\times$ median
	Feature dimensions	7D
Fusion	Voxel resolution	1.0 mm
	Outlier removal k	20 neighbors
	Outlier removal σ	2.0
Pose Graph	Voxel size	6.0 px
	Correspondence dist.	15.0 px

Table 3: Baseline implementation details.

Method	Model / Version	Hardware	Key Settings
ORB-SfM	OpenCV 4.13	CPU	3000 ORB features 5-pt essential No BA
COLMAP	3.8	CPU	SIFT, sequential Full BA
VGGT	official ckpt	GPU	1.2B params 18.4 GB window=8 frames
Reloc3R	official ckpt	GPU	DUST3R-based 12.1 GB Motion averaging
DUST3R	official ckpt	GPU	512-point maps 14.8 GB Global align.
MASt3R	official ckpt	GPU	Metric pointmaps 15.2 GB Global align.
MUST3R	official ckpt	GPU	8 keyframes, 224×224 memory network

Table 4: Pose error statistics for all methods against the independent `solvePnP` reference (first 50 frames of the chessboard sequence). ATE in mm; RPE-t in mm per frame pair. Stereo SGM shares the heightfield-ICP trajectory of Ours and is therefore identical to Ours on all pose metrics. Global rotation error is not reported: the sequence is translation-dominant (reference relative rotation $\approx 0.3^\circ$ per frame), so the rotational part of a position-fitted Sim3 alignment is ill-conditioned. **Bold**: best per column.

Method	n frames	ATE↓	RPE-t↓	Scale
Ours (heightfield-ICP)	50	3.89	3.10	1.11 ×
ORB-SfM	50	3.79	3.29	3.71×
COLMAP	41	20.10	3.28	8.86×
VGGT	50	3.02	2.56	90.9×
DUSt3R	50	30.78	2.58	647×
MASt3R	50	36.86	2.57	217×
MUSt3R	20	12.13	2.18	9.85×
Reloc3R	50	26.84	2.56	120×

Table 5: Complete PCA eigenvalue analysis. $\lambda_1 \geq \lambda_2 \geq \lambda_3$ are the eigenvalues of the point cloud covariance matrix. Extent shows the spatial range along each principal axis. Extents are in millimetres for the metric methods (Ours, Stereo SGM) and in each method’s native (non-metric) output units for the feed-forward and SfM baselines, whose scales are off by $3.7\times-647\times$ (Table 4); extents are therefore not physically comparable across rows.

Method	Points	λ_1	λ_2	λ_3	λ_2/λ_1	λ_3/λ_1	Extent
ORB-SfM	6,114	1.00	0.06	0.01	0.06	0.01	$135.3\times 59.8\times 199.8$
COLMAP	1,994	1.00	0.36	0.13	0.36	0.13	$4.0\times 6.9\times 7.5$
VGGT	324K	1.00	0.75	0.19	0.75	0.19	$2.5\times 1.5\times 2.5$
DUSt3R	229K	1.00	0.40	0.09	0.40	0.09	$0.37\times 0.28\times 0.24$
MASt3R	314K	1.00	0.82	0.22	0.82	0.22	$0.98\times 0.93\times 0.65$
MUSt3R	1,004K	1.00	0.19	0.06	0.19	0.06	$4.7\times 4.3\times 8.1$
Reloc3R	0	—	—	—	—	—	—
Stereo SGM	6,917	1.00	0.42	0.31	0.42	0.31	$78.3\times 119.6\times 103.3$
Ours	11,735	1.00	0.52	0.42	0.52	0.42	$191\times 200\times 193$

Table 6: Grid-step sensitivity analysis (poses held fixed). Smaller steps produce denser height-field meshes and denser fused clouds.

c_{cell} (px)	Mesh vertices	λ_3/λ_1	Points	Shape
2	57,280	0.389	62,410	Normal 3D
3	25,680	0.393	59,599	Normal 3D
4	14,400	0.328	25,346	Normal 3D
5	9,216	0.339	10,944	Normal 3D
7	4,784	0.318	9,104	Normal 3D
9	2,880	0.285	14,456	Normal 3D

Table 7: Search radius sensitivity (original pipeline artifacts). The reconstruction is identical for all tested radii.

r (px)	λ_3/λ_1	Points	Shape
10	0.429	11,735	Normal 3D
15	0.429	11,735	Normal 3D
20	0.429	11,735	Normal 3D
25	0.429	11,735	Normal 3D
30	0.429	11,735	Normal 3D
40	0.429	11,735	Normal 3D

Table 8: Learned stereo variants: in-domain (EndoNeRF validation) and zero-shot cross-dataset (SCARED) depth accuracy. Over-parameterised V3/V3s overfit the small training set; the pretrained V4 generalises best across datasets, while the lightweight V2 matches training-free SGM in-domain (AbsRel 0.0305 vs. 0.031).

Variant	Params	EndoNeRF (in-domain)			SCARED (zero-shot)		
		AbsRel↓	RMSE↓	δ_1 ↑	AbsRel↓	RMSE↓	δ_1 ↑
V2	4.76M	0.0305	6.96	0.9870	2.663	12.58	0.233
V2+	5.56M	0.0338	7.11	0.9865	2.955	13.89	0.209
V3	26.54M	0.1770	17.06	0.6324	2.782	13.10	0.202
V3s	23.05M	0.2082	17.11	0.6083	2.886	12.48	0.218
V4	5.01M	0.0668	8.53	0.9769	2.435	11.23	0.245

Table 9: Per-sequence depth accuracy and shape classification on the Stereo Laparoscopic Dataset. AbsRel and δ_1 computed after median-scale alignment against ground-truth depth. Ours and Stereo SGM share the same disparity source, so their depth metrics are identical by construction; the small shape differences arise from stochastic PCA subsampling. Shape: N=Normal 3D, F=Flat, C=Cone.

Seq	Ours		SGM		Mono		Ours	SGM	Mono
	AbsRel↓	δ_1 ↑	AbsRel↓	δ_1 ↑	AbsRel↓	δ_1 ↑	Shape	Shape	Shape
01	0.036	0.986	0.036	0.986	0.842	0.232	N	N	F
04	0.012	0.998	0.012	0.998	1.064	0.193	N	N	C
05	0.011	1.000	0.011	1.000	1.108	0.163	C	C	C
06	0.016	0.997	0.016	0.997	0.866	0.224	F	F	C
08	0.056	0.953	0.056	0.953	0.800	0.237	C	F	C
09	0.061	0.915	0.061	0.915	0.728	0.247	N	N	C
11	0.026	0.984	0.026	0.984	0.675	0.270	N	C	C
12	0.015	0.999	0.015	0.999	0.697	0.268	F	F	C
14	0.151	0.810	0.151	0.810	1.227	0.177	C	C	C
15	0.050	0.954	0.050	0.954	0.975	0.198	F	C	C
16	0.078	0.930	0.078	0.930	1.033	0.188	F	N	C
17	0.222	0.658	0.222	0.658	1.135	0.197	F	F	C
18	0.189	0.734	0.189	0.734	1.029	0.180	F	F	C
19	0.152	0.754	0.152	0.754	1.003	0.203	N	N	C
20	0.145	0.897	0.145	0.897	0.873	0.224	F	F	C
21	0.086	0.919	0.086	0.919	1.257	0.175	F	F	C
22	0.146	0.779	0.146	0.779	0.952	0.206	F	F	C
23	0.149	0.804	0.149	0.804	1.223	0.178	F	F	C
24	0.196	0.707	0.196	0.707	0.959	0.195	F	F	C
25	0.121	0.866	0.121	0.866	0.927	0.226	C	C	C
26	0.130	0.821	0.130	0.821	1.120	0.176	C	C	C
27	0.078	0.886	0.078	0.886	1.048	0.198	F	F	C
Avg	0.097	0.880	0.097	0.880	0.979	0.207	—	—	—

Table 10: Full per-session metrics of the proposed method on the newly captured sequences. λ_3/λ_1 : 3D shape ratio. ATE / RPE-t in mm against the cleaned PnP reference. Corner depth is evaluated at the 11×8 inner corners of the 3.0mm board with calibrated intrinsics after Sim3 alignment of the fused cloud and per-session affine alignment ($gt = a\text{-pred} + b$). V3’s low AbsRel is a local board-plane fit and does not contradict the Cone global shape. GT rot.: mean relative rotation per valid-frame step after flip removal (P1 also excludes steps $\geq 60^\circ$).

Seq	Frames	GT rot.	λ_3/λ_1	Shape	ATE↓	Scale	RPE-t↓	AbsRel↓	RMSE↓	$\delta_1\uparrow$
V1	535	2.4°	0.657	Normal 3D	27.5	0.47	14.0	0.065	15.4	0.978
V2	496	2.5°	0.221	Flat	58.4	0.63	13.8	0.050	13.5	0.993
P1	496	1.9°	0.119	Cone	55.2	0.35	22.4	0.070	19.8	0.991
V3	839	4.2°	0.053	Cone	38.8	0.099	63.7	0.020	4.79	1.000

Table 11: Per-frame position error (mm, mean) stratified by reference relative rotation per valid-frame step after flip removal. P1’s $>10^\circ$ bin has only 8 frames and may include residual flips. V3 is nearly flat across bins (32–40 mm), so its Cone collapse is not a per-step rotation effect.

Seq	0– 2°	2– 5°	5– 10°	$> 10^\circ$
P1	40.5	43.6	32.0	90.9
V1	22.7	28.0	30.9	36.5
V2	51.0	48.0	51.7	62.5
V3	31.6	38.5	39.6	39.3

Table 12: Board occupancy and 12×9 –3 mm working distance. Secondary-board distances come from the same labelled CSV and are diagnostic only.

Seq	Frames	12×9	9×6	9×8	Dist. mean	median	p95 (mm)
V1	535	127	3	0	187	190	220
V2	496	135	5	1	199	196	235
P1	496	129	19	1	216	212	257
V3	839	98	0	1	177	180	202

Table 13: Branch-selection accuracy (fraction of pairs selecting the reference-consistent branch) and median translation-direction agreement $|\hat{\mathbf{t}}^\top \mathbf{t}_{\text{ref}}|$. The veto with a faithful normal matches or exceeds the cheirality heuristic at every stride, with the largest margin at stride 8 (73.8% vs. 69.0%), where the baseline is largest and the two branches are most separable. The heuristic mis-selects 24–33% of pairs. Translation agreement is limited by decomposition noise at the 2–20 mm per-pair displacements of this sequence and is reported for completeness.

Stride	Rule	n	Branch acc. $\uparrow$	Median $ \hat{\mathbf{t}}^\top \mathbf{t}_{\text{ref}} $
1	Veto (true normal)	49	0.755	0.564
	Veto (rig prior)	49	0.490	0.393
	Cheirality heuristic	49	0.755	0.564
2	Veto (true normal)	48	0.667	0.423
	Veto (rig prior)	48	0.646	0.396
	Cheirality heuristic	48	0.667	0.423
4	Veto (true normal)	46	0.696	0.327
	Veto (rig prior)	46	0.609	0.396
	Cheirality heuristic	46	0.696	0.327
8	Veto (true normal)	42	0.738	0.374
	Veto (rig prior)	42	0.595	0.243
	Cheirality heuristic	42	0.690	0.374